

Digha Tide Predictor 2.0

A Scientific Roadmap from an Empirical Prototype to a Research-Grade Tide Prediction System

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The present Digha Tide Predictor is an experimental harmonic-analysis project built from a short source dataset. Its next stage should focus not merely on adding more code, but on improving the quality of the data, the physical model, the numerical estimation, and the validation framework. The following ten developments define a practical roadmap for Tide Predictor 2.0.

1. The Data Foundation — From 38 Extrema to Continuous Ocean Observations

Acquire genuine Digha tide-gauge observations at ideally 1-minute or 5-minute intervals. Thirty days should be the minimum serious starting point; several months, and ultimately a full year, would be much stronger. This changes the project from fitting a handful of turning points to analysing the complete tidal waveform.

2. The Station Identity — Location, Time and Vertical Datum

Record the exact tide-gauge station, coordinates, time standard, units and vertical datum/chart datum. A scientifically useful prediction must know exactly what the reported water height means and what reference level it is measured from.

3. The Harmonic Spectrum — Expanding Beyond Six Constituents

Move gradually beyond M2, S2, N2, K1, O1 and M4. Add constituents such as P1, K2, Q1, MS4 and others only when the available record is long enough to identify them reliably. The aim is a richer physical representation without creating an over-parameterised model.

4. Astronomical Precision — Full Sun-Moon Forcing

Implement the astronomical arguments of the tidal constituents rather than relying only on fixed constituent speeds. This will allow the model to represent the changing astronomical configuration of the Earth-Moon-Sun system at the prediction date.

5. Nodal Dynamics — Correcting the Long-Period Lunar Effects

Introduce nodal amplitude and phase corrections. These account for slow variations associated with lunar orbital geometry and are an important part of professional harmonic tide prediction.

6. Numerical Engine 2.0 — Robust Statistical Estimation

Upgrade the fitting engine with weighted least squares, QR or singular-value decomposition, systematic outlier detection, missing-data handling and numerical conditioning diagnostics. The goal is a stable estimator that remains reliable when the dataset becomes large and the constituent set becomes more complex.

7. Validation — Testing the Future, Not Memorising the Past

Separate calibration data from genuinely unseen observations. For example, fit the model on one period and test it on a later month. This provides a real measure of predictive performance rather than simply measuring how well the model reproduces the data it was given.

8. Dual Accuracy — Height Error and Timing Error

Report at least two separate classes of accuracy: water-level error in metres and high/low-water timing error in minutes. A prediction can have a small height error while still placing a high or low tide noticeably early or late, so both must be measured independently.

9. Beyond Astronomy — Meteorological and Oceanographic Residuals

After the astronomical tide is modelled, investigate the remaining residual water level using atmospheric pressure, wind, rainfall, river discharge and storm information. This would move the system toward a combined astronomical-tide-plus-weather model.

10. Tide Predictor 2.0 — A Scientific Prediction Platform

Keep the current interactive JavaScript interface, but turn it into the front end of a complete analysis system. Add observed-versus-predicted plots, uncertainty bands, constituent diagnostics, validation statistics, downloadable predictions and eventually a research-grade numerical backend.

The central transition is simple:

38 tide extrema over 10 days → continuous observations over months/year → full harmonic analysis → independent validation → Tide Predictor 2.0

The present program is the foundation. The next version should preserve its educational transparency while progressively introducing the data and mathematical machinery required for a scientifically defensible prediction system.

Until the next stage of the project — we will build it from the data upward.